

Programming and Data Structures Laboratory (CS19003)

Lab Test-I (Section 6)

Department of CSE, IIT Kharagpur

11th February, 2026

General instructions to be followed strictly:

- Name your file as `<roll_no>_<assignment_no>_<question_no>`. For example, if your roll number is 14CS10001 and you are submitting assignment 3 question number (a), then name your file as 14CS10001_3_a.c. Write your name, roll number, and assignment number at the beginning of your program.
 - Use proper indentation in your code and comment. **Marks may be deducted without proper indentation and comments.**
 - Submitting the **correct file** is your responsibility, any request of re-submission after the lab will **never** be entertained.
 - **Zero tolerance** policy will be strictly followed for **plagiarism**. The marks will be put straight zero if such cases are identified.
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Question-1 (10 points)

You are given a two-digit positive integer N such that $10 \leq N \leq 99$. Let the tens digit of the number be a and the units digit be b . Extract these digits and analyze the number according to the conditions described below.

The number must be classified according to the **first condition that is satisfied**, in the order listed below: If the sum of the digits $a + b$ is a prime number, classify the number as **Type P**. If this condition is not satisfied, check whether one digit divides the other exactly, with neither digit being zero; if so, classify the number as **Type D**. If neither of the previous conditions applies, check whether the value $a^2 + b^2$ is divisible by 10; if this condition holds, classify the number as **Type S**.

If none of the above conditions are satisfied, classify the number as **Type O (Other)**. Your task is to read the input number, determine the appropriate classification, and print only the corresponding type (P, D, S, or O).

Example:

(Input)

Enter the number: 29

(Output)

The classification is: Type P

Question-2 (15 points)

You are given a positive integer N containing at least three digits. Starting from the least significant digit, process the digits of the number one by one towards the most significant digit. While processing the digits, consider only contiguous sequences of digits formed during this right-to-left traversal.

A contiguous digit sequence is said to form a *valid alternating run* if the relationship between every pair of consecutive digits alternates between *strictly increasing* and *strictly decreasing*. That is, if one comparison shows that a digit is greater than the previous digit, the next comparison must show a digit smaller than the previous one, and this alternation must continue throughout the sequence. If two consecutive digits are equal, the alternating run is immediately broken.

For example, consider the number 73926184. When processed from right to left, the digits are examined in the order 4, 8, 1, 6, 2, 9, 3, 7. The digit sequence 1, 6, 2, 9, 3 forms a valid alternating run because the comparisons alternate as follows: $6 > 1$, $2 < 6$, $9 > 2$, and $3 < 9$. The length of this run is 5, which is the maximum possible for this number.

Your task is to determine the maximum length of any valid alternating run present in the given number. If no alternating run of length greater than 1 exists, print 1.

Example:

(Input)

Enter the number: 73926184

(Output)

Maximum length of valid alternating run is 5

Question-3 (10 points)

Write a program that takes as input a sequence of 1, 2, and 3 from the user and stores the sequence in an integer array. First ask the user the number of elements in the sequence, and then ask the user to enter the elements one by one. Your program should check whether every input element is 1 or 2 or 3; if the user inputs any other number, the user should be asked to input that element again. You can assume that the maximum length of the input sequence is 100.

Then, your program should identify and print the starting indices of the *longest consecutive sequence* of 1, 2, and 3 in the given sequence. If there are multiple longest consecutive sequences (of equal length) of an element, you can print the starting index for any one of the longest sequences. In case the input sequence does not contain any of these elements at all, the starting index for that element should be printed as -1.

Example:

(Input)

Input sequence: 1 2 2 2 1 3 3 3 1 1

(Output)

Start of the longest consecutive sequence of 1: 8

Start of the longest consecutive sequence of 2: 1

Start of the longest consecutive sequence of 3: 5

Question-4 (15 points)

Consider an array A of size n which is filled with real numbers. The size n of the array and the real numbers should be taken as input from the user. Write a C-program that prints the sum of all unique elements in the array, and also prints the unique numbers in the format shown below. An unique element appears only once in the array. You can take the maximum possible size of the array to be 30. Some **example** arrays and outputs are given below. **Note** that you are not allowed to use any additional array other than A inside your program.

Example 1: Size of the array with 8 elements [1 3 2 4 5 3 6 2] will have the output:

Sum of unique elements (1 4 5 6) is 16

Example 2: Size of the array with 4 elements [5 5 5 5] will have the output:

Sum of unique elements () is 0

The questions end here.

The next few pages are added extra. You can use these, if for any question you need extra space.
